

FYP 2024/2025 (Telecom and Power)

1-Dr. Hussein Al Haj Hassan

Project1: Using UAVs for enhanced wireless communication in smart metering networks.

Project2: Identification of products in retail stores.

2-Dr. Ahmad Raad

Project1: Fixed wireless access (FWA) for rural areas in Lebanon.

3-Dr. Mohamad Aoude

Project1: Quantum-resistant cryptography for next-generation secure communications.

Project2: Generative adversarial networks for synthetic cybersecurity threat simulation.

Project3: AI-driven resource allocation for 6G ultra-reliable low-latency communications.

Project4: Quantum key distribution integration in next-generation optical networks.

Project5: AI-Enhanced network slicing for secure and efficient 6G telecommunications.

Project6: AI-powered quality of service (QoS) management in wireless networks (appears twice).

Project7: Quantum key distribution (QKD) for secure satellite communication.

Project8: Explainable AI (XAI) for decision support systems.

4-Dr. Rima Assaf

Project1: Autonomous navigation system: path planning, obstacle avoidance, and GPS localization.

Project2: Comprehensive survey and implementation of text classification techniques.

Project3: Comprehensive survey and implementation of speaker recognition techniques.

5-Dr. Hiba Haj Chhade

Project1: Wind energy fed electric vehicle charger using MPPT at variable wind speed.

Project2: Performance analysis of ANN based DC to DC converter.

6-Dr. Hayat Nasser

Project1: Firearm detection using deep learning.

Project2: Gesture recognition using neural networks.

Project3: Face skin analysis using machine learning.

7-Dr. Rida EL Chall

Project1: Recommender system based on graph neural network.

Project2: Face image quality assessment using deep learning.

Project3: Smart agriculture: crop yield and disease detection using deep learning.

8-Dr. Bilal Komati

Project1: Design and development of an AI educational robot.

Project2: Self-reconfigurable robot with programmable magnets.

Project3: Design of a novel voltage amplifier for piezoelectric actuators.

Project4: Design and implementation of a modular IoT platform for localization and data transmission.

Project5: Design of a ground station for communication with LEO satellites.

Project6: Application of LoRa technology for remote sensing in farms.

9-Dr. Youssef Harkouss and Dr. Mohamad Aoude

Project1: AI-based traffic control system for traffic jam prevention.

10-Dr. Youssef Harkouss

Project1: Adapting SLAM systems for multi-fisheye camera inputs in autonomous vehicles.

11-Dr. Youssef Harkouss and Eng. Mohamad Halabi

Project1: Advanced sign language communication system for real-time interaction and accessibility

Project2: Deep learning-based system for audio restoration and enhancement.

12-Dr. Amjad EL Hajjar

Project1: Data analysis and machine learning for detecting and classifying malicious URLs.

Project2: Survey and application of image template-matching methods.

Project3: Survey and application of image super-resolution methods.

Project4: Prototype project for using moodle LMS for the faculty of engineering.

Project5: Data analysis and machine learning methods for Forex trading.

Project6: Semantic search using supplementary text.

13-Dr. Abed Ellatif Samhat and Dr. Ali Ghandour

Project1: Data processing pipeline for raw satellite imagery.

Project2: Super-resolution of satellite imagery.

Project3: XAI for earth observation models.

14-Dr. Lana Damaj

Project1: Mutual coupling reduction in array antennas.

Project2: Design and optimization of multi-band energy-efficient antennas for advanced 5G applications.

15-Dr. Abir El Abed

Project1: Automated detection for neonatal jaundice using image processing techniques and machine learning algorithms.

Project2: Machine learning-based identification and classification of artifacts in vital signs during MRI scans.

16-Dr. Jihad Al Sahili

Project1: Designing and integrating the onboard controller of an underwater remotely operated vehicle ROV (Co-supervisor: Eng. Ali Nehmeh).

Project2: Open source motion capture for autonomous drones (Co-supervisor: Eng. Hussein Mazeh).

Project3: Implementing of a ground control station (GCS) application of an underwater remotely operated vehicle ROV (Co-supervisor: Eng. Ali Nehmeh).